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In[*]:= (*ESTELSCALIBRATGE*)

In[*]:= ECCordEct = {28.815969`, 29.608397`, 29.141317`,
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In[*]:= ECDec = {36.868473`, 38.342148`, 37.630542`,
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In[*]:= ECBrillG = {783.5690271`, 1754.341773`, 19545.38855`, 4.571231439`*^6,
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In[*]:= ECBrillB = {135.3277122`, 446.4417029`, 8117.457561`, 2.143295927`*^6,
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In[*]:= ECMagApB = {20.01`, 18.71`, 15.56`, 9.51`, 13.82`, 12.53`, 9.553`, 10.98`};

In[*]:= ECBrillR = {1027.386948`, 1943.098866`, 16379.77599`,
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In[*]:= ECTeff = {3238.4`, 3664.2`, 4548.5`, 5052.8`, 5525.9`, 6024.7`, 6505.4`, 7244.9`};

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In[*]:= (*CUMULTXT*)

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  17731.58135`, 7136.434945`, 8201.333819`, 10895.33521`, 13481.4608`, 12858.00206`,
  7526.713177`, 7732.148741`, 8594.952612`, 26032.14128`, 11142.83562`, 16521.99204`,
  11819.39891`, 13564.96597`, 15366.41231`, 717.5376924`, 14633.75367`, 12158.1313`,
  16379.77599`, 14202.84691`, 1995.273145`, 32600.68413`, 17196.55098`, 16910.82729`,
  17616.91512`, 28505.92108`, 15977.3936`, 16252.26018`, 40494.77105`, 5415.679415`,
  17856.34689`, 3.381534929`^6, 21288.88982`, 4.047847252`^6, 4.518032647`^6,
  5.056090334`^6, 5.182899057`^6, 4.998177026`^6, 25734.1463`, 4.702895429`^6,
  22753.46498`, 4.515488847`^6, 2.800977911`^6, 22702.15179`, 20863.68549`,
  22586.74916`, 24494.59144`, 2.883075897`^6, 3.455541397`^6, 1.486097113`^6,
  3.423767975`^6, 32718.07858`, 40737.11993`, 4019.37875`, 33324.15577`, 83636.1656`,
  35000.02976`, 75340.04282`, 39814.69796`, 38448.71376`, 45880.30007`, 7582.010667`,
  46132.03612`, 9763.520412`, 46895.45306`, 4.459527824`^6, 54812.55709`,
  33384.3655`, 57100.19329`, 60744.45975`, 58698.5568`, 69972.41396`, 93881.18775`,
  35632.27431`, 98150.48758`, 91907.55499`, 113448.7932`, 74219.79513`, 96316.46958`,
  100420.1242`, 59094.71685`, 65222.80819`, 72277.22389`, 104972.9856`, 242776.5092`,
  116649.1121`, 166639.3846`, 95885.003`, 130143.2511`, 107625.9521`, 120507.5747`,
  127478.0948`, 164692.9192`, 429.3385974`, 133189.8367`, 118599.9254`, 141728.8853`,
  148835.677`, 100460.7397`, 147013.1842`, 154740.8158`, 165339.6452`, 300869.7242`,
  168981.0515`, 163197.0411`, 350979.0701`, 122506.4203`, 153602.4887`, 256630.8412`,
  190147.0244`, 228429.0963`, 431414.61`, 1.527301278`^6, 1.09950912`^6,
  1.254377497`^6, 218477.6893`, 1.086330573`^6, 1.195814733`^6, 1.059516097`^6,
  2.153515582`^6, 479891.3415`, 1.33868743`^6, 1.219622668`^6, 530566.8993`,
  272516.3523`, 256036.2494`, 1.167336757`^6, 410406.7301`, 262862.1389`,
  1.417225056`^6, 248696.8986`, 273385.519`, 324869.214`, 412471.3804`, 294126.609`,
  1.041994628`^6, 283510.169`, 815638.6188`, 321483.937`, 1.759869215`^6,
  335955.1024`, 365893.7701`, 357674.8674`, 306419.7294`, 1.115468237`^6,
  393346.6764`, 826635.6016`, 817109.5722`, 488236.4053`, 744450.9594`,
  1.135565458`^6, 398092.0107`, 435656.5927`, 605924.3822`, 490687.1577`,
  505341.6868`, 735412.1549`, 787452.0931`, 783407.7333`, 589739.6796`, 726655.0853`,
  629604.1112`, 765169.6727`, 698895.8579`, 822325.1452`, 463632.7354`, 509257.9389` };

In[*]:= (*-----*)

In[*]:= (*HYADESTXT*)

```

```
ln[*]:= HYCordEct = {67.48275769215242`, 67.16468609875307`, 67.49102855410932`,
    67.20174925382867`, 67.04582690253649`, 67.25111621738195`,
    67.79637749253605`, 66.81983230907242`, 67.37958519537817`, 66.6676726168485`,
    67.64062132435598`, 67.12038862642493`, 66.77789108447449`, 66.77717673131863`,
    67.69551967147464`, 66.57982992496277`, 66.92021718932686`, 67.87161425364759`,
    68.0336649745617`, 67.37695672528574`, 68.40865079618047`, 67.38219601870492`,
    67.2190620079301`, 68.48680175601453`, 68.09949961823423`, 67.16608051054888`,
    66.30695338557209`, 68.20929931916466`, 67.96612153925633`, 66.31936239676666`,
    66.11855100326646`, 67.66251575060892`, 66.09326952772372`, 67.53632737172786`,
    66.59090387218913`, 68.24817726690902`, 67.93658293586253`, 67.9373303656096`,
    65.99690955661914`, 66.3729376285106`, 66.40598637273915`, 66.35479242496689`,
    66.02741879597104`, 66.02452121402943`, 66.90003194033417`, 65.98260314349544`,
    67.96911967367434`, 65.85493019594335`, 68.63451075082207`, 66.20070596676636`,
    66.99617799986078`, 66.11822759101871`, 66.52490368958763`, 67.30193646682095`,
    66.07110065917192`, 67.25475392693583`, 68.48159725897497`, 65.73422490902678`,
    65.92951789088579`, 68.64759095859309`, 66.9971739315349`, 69.38416312823124`,
    69.38447123432836`, 65.85591280295498`, 65.39849909000245`, 67.93265109428891`,
    69.02238770364617`, 69.1701171674695`, 67.81866462842426`, 65.80253436758252`,
    68.42892746109855`, 68.49438499286096`, 68.86895204897276`, 69.63189917287164`,
    68.66689038204488`, 66.52007676867467`, 69.60440899071648`, 65.66541825180812`,
    66.46067898955343`, 68.15820465974114`, 67.6415947161492`, 67.15467370962931`,
    68.42516726156403`, 68.42662834627303`, 68.17103556840848`, 69.61596544100632`,
    66.31113667576857`, 67.6780424615967`, 69.5444304347343`, 68.4696324759209`,
    68.4626315976831`, 69.16277554836863`, 65.75678797400016`, 66.23851281426906`,
    66.12800628155945`, 64.97907765724916`, 65.38496942725972` };
```

```

In[*]:= HYDec = {16.913911822491304`, 16.969906003224942`, 16.67271469531208`,
  17.285341899758865`, 16.47079736733054`, 16.346213879024717`, 16.39576252180493`,
  17.241711375171782`, 16.244692315734465`, 16.74677258632566`, 16.193900711881753`,
  17.695791947186837`, 16.429579986614307`, 16.430092002554506`, 16.148532608296946`,
  17.05047474838996`, 16.248607862645795`, 17.718518754515433`, 17.6643440638911`,
  17.863022648846673`, 16.762386244728727`, 17.89303687476587`, 15.981472635522984`,
  16.86893699017762`, 17.750589386258458`, 15.870784455320392`, 17.268076545714347`,
  16.00568312875324`, 15.851471350184573`, 16.3022336834695`, 16.886053964801455`,
  15.691769659062853`, 17.07883604085962`, 15.637736095425185`, 18.00015324250052`,
  15.818854771000632`, 15.630149907181586`, 15.629243097689292`, 16.721465710411213`,
  17.927768735352878`, 15.940907539697319`, 17.91521008711174`, 17.4442235658507`,
  17.443985101312077`, 15.589085472787543`, 16.354077059105514`, 15.49937349658422`,
  16.777147465731876`, 15.827466076708385`, 15.874649203665436`, 18.500100518411095`,
  15.88417682534391`, 15.524230392191061`, 15.273804206793624`, 18.00278420332121`,
  18.67354031760568`, 18.38062040059367`, 17.542390757517502`, 15.880803450919558`,
  15.504503855092008`, 18.75895052871627`, 16.75879068634628`, 16.758565709467636`,
  15.763005166817576`, 16.894192274532255`, 15.041007895036627`, 15.683897520258279`,
  15.869204695827205`, 15.00322280399349`, 15.712784182546748`, 18.766254549927957`,
  15.16352719505778`, 15.399151275857333`, 17.04055509824066`, 15.208964911975178`,
  15.041285019426095`, 17.54228778068461`, 18.269179460705352`, 15.002464972130216`,
  19.046525581821314`, 14.747976746494444`, 19.180273317307044`, 19.01388394635447`,
  19.013283769225044`, 19.113332347757908`, 16.00290935094271`, 18.973455972926324`,
  14.66142954817377`, 15.821382952558132`, 14.850761740625991`, 14.844319676275646`,
  18.615579882548317`, 15.228121233684698`, 19.041854836257855`,
  18.986786982710726`, 16.522417532947376`, 18.41744025238449` };

```



```

In[*]:= HYParallax = {22.72646779815639`, 20.707493378225784`, 21.918833298371936`,
    21.5489076022758`, 23.619650052576446`, 21.632519360189345`, 20.316503143723228`,
    21.81444620546993`, 21.81806190139089`, 21.852128841107575`, 22.18747294665825`,
    20.91843408231149`, 21.7524975456685`, 21.66732573514696`, 22.452802624244807`,
    21.204050746536424`, 18.8055441535833`, 22.235741889084355`, 21.07527631220369`,
    21.438554423234333`, 20.61556113871976`, 22.00647700341757`, 22.15827839888896`,
    21.48256636168679`, 23.39160255904337`, 21.81251596251546`, 19.31283635339333`,
    21.986514300716408`, 21.00505488455235`, 21.250239265348167`, 21.781982400489085`,
    21.86843191355519`, 19.95899563246898`, 21.420137280794197`, 19.427532599125378`,
    21.35731378098158`, 21.928109289904448`, 21.9411429349474`, 21.74597706160512`,
    21.086509524006694`, 21.10764322783181`, 21.447339313427232`, 21.645976040714597`,
    21.61924104437301`, 21.284016509945314`, 22.088260260658192`, 21.871807832755163`,
    20.79504603398395`, 19.196267295270147`, 20.678332758167166`, 19.76046789154403`,
    20.427173906644025`, 23.521027428410385`, 21.451662078172305`, 21.496896698223075`,
    23.141968871394663`, 21.587850129123968`, 20.30919733178576`, 22.4770888204994`,
    21.668969928652356`, 19.12817304877517`, 21.410085237889614`, 21.28243624077011`,
    24.13445982542496`, 24.3377306899844`, 19.653211597272005`, 19.993054066507124`,
    21.607898747664954`, 21.05318989100504`, 21.566208065928656`, 24.168416936596586`,
    20.64126832294862`, 21.717186315251215`, 22.85331654281413`, 21.07507415226267`,
    20.8390589991997`, 21.502834748637135`, 24.013842432491053`, 21.39808326312012`,
    23.275164118655212`, 20.132587476239593`, 22.36535445562801`, 20.871118754097125`,
    20.302634235467437`, 19.02878834888202`, 20.892995227960203`, 18.569708027172418`,
    24.44786660420823`, 21.01838989752342`, 21.94460899596352`, 21.635396574818312`,
    23.802781584110832`, 20.706527940234576`, 21.338379524598544`,
    20.49866479939148`, 21.725333783131415`, 20.885733460424145` };

In[*]:= HYMagApG = {11.965812`, 14.05441`, 8.859189`, 7.7025585`, 10.505475`, 13.074519`,
    13.791429`, 12.13662`, 9.932404`, 7.950683`, 4.752029`, 11.360385`, 16.60488`,
    16.533167`, 6.406904`, 16.338469`, 14.2993765`, 11.184438`, 13.979582`, 6.797135`,
    9.15698`, 8.7407255`, 11.251079`, 13.742438`, 13.353131`, 3.3779078`, 11.998245`,
    8.728549`, 5.928877`, 14.998649`, 7.6696334`, 5.4175873`, 6.855921`, 5.6056676`,
    14.301552`, 8.475605`, 13.436778`, 11.270382`, 11.733229`, 4.295662`, 6.341617`,
    10.60706`, 14.583092`, 4.7558556`, 7.289236`, 14.329463`, 10.49808`, 5.573977`,
    9.416291`, 11.324224`, 9.7873955`, 13.778803`, 7.3494782`, 11.554099`, 9.650242`,
    12.25473`, 14.972231`, 3.516044`, 13.549859`, 7.782416`, 12.942235`, 14.560182`,
    14.52253`, 10.021945`, 13.948855`, 11.476107`, 9.142798`, 6.5345774`, 16.090866`,
    14.3163395`, 13.956485`, 8.2856655`, 14.505741`, 13.618415`, 11.549942`, 11.470464`,
    9.699967`, 12.078186`, 14.749543`, 13.744849`, 13.6755705`, 3.2804532`, 10.271567`,
    18.51686`, 10.12441`, 17.021824`, 11.957183`, 15.196015`, 14.388778`, 15.195518`,
    4.6015615`, 12.300638`, 15.144566`, 5.881098`, 14.982163`, 7.0219374`, 6.7013855` };
    
```

```
In[*]:= HYMagApB = {13.101174`, 13.947589`, 9.285927`, 8.002354`, 11.301343`, 14.537198`,
  15.336586`, 13.280433`, 10.553728`, 8.277002`, 4.8255715`, 12.490327`, 18.97145`,
  18.870968`, 6.6114206`, 18.545973`, 15.886161`, 12.11163`, 15.532152`, 7.0180893`,
  9.601489`, 9.148271`, 12.190303`, 15.293826`, 14.032649`, 3.574795`, 13.481207`,
  9.129763`, 6.094176`, 16.785208`, 7.962704`, 5.5338874`, 7.0739436`, 5.71062`,
  16.138481`, 8.842474`, 14.924236`, 12.222652`, 12.765127`, 4.3194156`, 6.5708666`,
  11.361621`, 15.820655`, 4.8421674`, 7.5509853`, 14.252101`, 11.237315`, 5.7164`,
  9.876807`, 11.849495`, 10.3645735`, 15.238744`, 7.62215`, 12.532283`, 10.201229`,
  13.527419`, 16.758175`, 4.063456`, 15.040574`, 8.089807`, 14.446476`, 15.731204`,
  15.961537`, 10.702273`, 15.735156`, 12.463258`, 9.573993`, 6.749979`, 18.275764`,
  16.050575`, 15.654839`, 8.699686`, 16.336432`, 15.206414`, 12.57892`, 12.402906`,
  10.367044`, 13.253375`, 16.512846`, 15.3451805`, 14.962966`, 3.898934`, 10.938479`,
  19.44101`, 10.713506`, 19.451487`, 12.925601`, 17.167006`, 16.064022`, 17.019016`,
  4.7148905`, 13.52421`, 16.95227`, 6.0735607`, 16.828726`, 7.0094604`, 6.9139585`};
```

```
In[*]:= HYMagApR = {10.901797`, 14.286898`, 8.266227`, 7.232789`, 9.635134`, 11.891975`,
  12.5815735`, 11.069566`, 9.181026`, 7.457401`, 4.592621`, 10.255144`, 15.199033`,
  15.144875`, 6.0271587`, 14.968907`, 13.080908`, 10.233044`, 12.694855`, 6.418047`,
  8.548466`, 8.164915`, 10.293494`, 12.532885`, 12.423298`, 3.1754673`, 10.796188`,
  8.043712`, 5.6289883`, 13.727402`, 7.1908584`, 5.20475`, 6.4828286`, 5.2629566`,
  13.015605`, 7.9380226`, 12.22202`, 10.307268`, 10.714438`, 4.2342367`, 5.926525`,
  9.764299`, 13.222724`, 4.6124096`, 6.8615117`, 14.504852`, 9.66431`, 5.3160176`,
  8.79175`, 9.926568`, 9.064594`, 12.464239`, 6.9091825`, 10.5701685`, 8.953043`,
  11.13354`, 13.700722`, 2.8761232`, 12.357251`, 7.3038206`, 11.743436`, 12.790547`,
  12.79442`, 9.216966`, 12.670804`, 10.437661`, 8.545767`, 6.1650352`, 14.732483`,
  13.059443`, 12.703316`, 7.6744475`, 13.204195`, 12.397598`, 10.487623`, 10.491246`,
  8.898517`, 10.997761`, 13.4863405`, 12.519701`, 12.2476225`, 2.6619565`, 9.488083`,
  16.561226`, 9.398879`, 15.608153`, 10.876818`, 13.884905`, 13.145536`, 13.89428`,
  4.3678617`, 11.201944`, 13.869616`, 5.5433846`, 13.622684`, 6.300481`, 6.3316135`};
```

```
In[*]:= (*-----
                                     -----*)
```

```
In[*]:= (*APARTADO 1*)
```

```
In[*]:= CalibCg = ECMagApG + 2.5 * Log10[ECBrillG];
                                     |logaritmo en base 10
```

```
In[*]:= MedCg = Mean[CalibCg]
                                     |media
```

```
Out[*]:=
25.683
```

```
In[*]:= StandardDeviation[CalibCg]
                                     |desviación estándar
```

```
Out[*]:=
0.00337983
```

```
In[*]:= CalibCb = ECMagApB + 2.5 * Log10[ECBrillB];
                                     |logaritmo en base 10
```

```
In[*]:= MedCb = Mean[CalibCb]
                                     |media
```

```
Out[*]:=
25.3349
```

```

In[*]:= StandardDeviation[CalibCb]
         [desviación estándar]

Out[*]=
0.00321375

In[*]:= CalibCr = ECMagApR + 2.5 * Log10[ECBrillR];
         [logaritmo en base 10]

In[*]:= MedCr = Mean[CalibCr]
         [media]

Out[*]=
24.7447

In[*]:= StandardDeviation[CalibCr]
         [desviación estándar]

Out[*]=
0.00339145

In[*]:= (*-----*)
         -----*)

In[*]:= (*APARTADO 2*)

In[*]:= Gcum = MedCg - 2.5 * Log10[CUMBrillG];
         [logaritmo en base 10]

In[*]:= Bcum = MedCb - 2.5 * Log10[CUMBrillB];
         [logaritmo en base 10]

In[*]:= Rcum = MedCr - 2.5 * Log10[CUMBrillR];
         [logaritmo en base 10]

In[*]:= (*-----*)
         -----*)

In[*]:= (*APARTADO 3*)

In[*]:= BRcum = Bcum - Rcum;

In[*]:= BRcumIntrins = BRcum - 0.044;

In[*]:= (*-----*)
         -----*)

In[*]:= (*APARTADO 4*)

In[*]:= pointsHR = Transpose[{BRcumIntrins, Gcum}];
         [transposición]

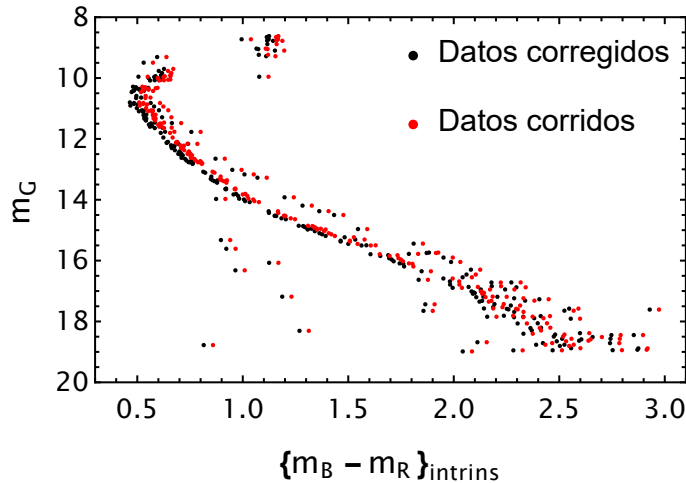
```

```
In[*]:= L1 = ListPlot[pointsHR, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  representación de lista | estilo base | familia de tipo de letra | tamaño de tipo de letra
  PlotRange → {{0.3, 3.1}, {8, 20}}, Frame → True,
  rango de representación | marco | verdadero
  FrameLabel → {Style[Subscript[{(Subscript[m, B] - Subscript[m, R])}], intrins],
  etiqueta de marco | estilo | subíndice | subíndice | subíndice
    16, Bold], Style[Subscript[m, G], 16, Bold]},
  negrita | estilo | subíndice | negrita
  PlotStyle → {Black, Thickness[0.003]}, ScalingFunctions → "Reverse",
  estilo de repre... | negro | grosor | funciones de escalado | invierte orden
  PlotLegends → Placed[PointLegend[{Black}, {"Datos corregidos"}], {Right, Top}]]];
  leyendas de rep... | colocado | punto leyenda | negro | derecha | arriba
```

```
In[*]:= L2 = ListPlot[Transpose[{BRCum, Gcum}],
  represent... | transposición
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  estilo base | familia de tipo de letra | tamaño de tipo de letra
  PlotRange → {{0.4, 3.0}, {8, 20}}, Frame → True,
  rango de representación | marco | verdadero
  FrameLabel → {Style[Subscript[{(Subscript[m, B] - Subscript[m, R])}], intrins],
  etiqueta de marco | estilo | subíndice | subíndice | subíndice
    16, Bold], Style[Subscript[m, G], 16, Bold]},
  negrita | estilo | subíndice | negrita
  PlotStyle → {Red, Thickness[0.003]}, ScalingFunctions → "Reverse",
  estilo de repre... | rojo | grosor | funciones de escalado | invierte orden
  PlotLegends → Placed[PointLegend[{Red}, {"Datos corridos"}], {Right, Top}]]];
  leyendas de rep... | colocado | punto leyenda | rojo | derecha | arriba
```

```
In[*]:= Show[L1, L2]
  muestra
```

Out[*]=



```
In[*]:= (*-----*)
  -----*)
```

```
In[*]:= (*APARTADO 5*)
```

```
In[*]:= BRcalib = ECMagApB - ECMagApR;
```

```
In[*]:= BRcalibIntrins = BRcalib - 0.044;
```

```
In[*]:= datatofit = Transpose[{BRcalibIntrins, ECTeff}];
```

transposición

```
In[*]:= fitTBR = NonlinearModelFit[datatofit, A * (x^(-b)), {A, b}, x]
```

ajusta a modelo no lineal

```
Out[*]=
```

FittedModel[$\frac{5140.}{x^{0.442}}$]

```
In[*]:= fitTBR[{"BestFit", "ParameterTable"}]
```

```
Out[*]=
```

		Estimate	Standard Error	t-Statistic	P-Value
$\frac{5138.62}{x^{0.441818}}$	A	5138.62	21.7626	236.121	3.89375×10^{-13}
	b	0.441818	0.00766831	57.616	1.83647×10^{-9}

```
In[*]:= plotfitTBR =
```

```
Plot[fitTBR[x], {x, 0, 15}, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
```

representación gráfica

estilo base

familia de tipo de letra

tamaño de tipo de letra

```
PlotStyle -> {Blend[{Blue, White}], Thick},
```

estilo de representación

mezcla

azul

blanco

grueso

```
PlotRange -> {{0.3, 3.2}, {3000, 7500}}, Frame -> True,
```

rango de representación

marco

verdadero

```
FrameLabel -> {Style[Subscript[{(Subscript[m, B] - Subscript[m, R])}], intrins],
```

etiqueta de marco

estilo

subíndice

subíndice

subíndice

```
Bold, 16], Style["Te / K", 16, Bold] }];
```

negrita

estilo

negrita

```
In[*]:= plotdataoffit = ListPlot[datatofit, PlotStyle -> {Black, Thickness[0.003]}];
```

representación de lista

estilo de representación

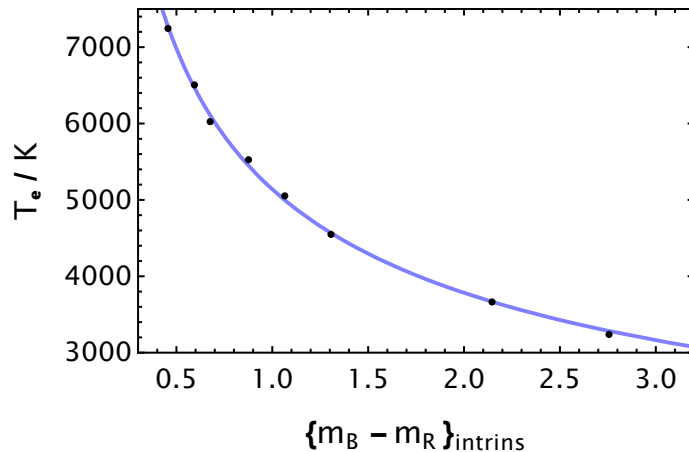
negro

grosor

```
In[*]:= Show[plotfitTBR, plotdataoffit]
```

muestra

```
Out[*]=
```



```
In[*]:= (*-----*)
```

-----*)

```
In[*]:= (*APARTADO 6*)
```

```
In[*]:= Teffcum = 5138.619786890803 * BRcumIntrins^(-0.4418177507509579);
```

```
In[*]:= (*-----*)
```

-----*)

```

In[*]:= (*APARTADO 7*)

In[*]:= ECMagApG[[8]]
Out[*]=
10.8

In[*]:= ECMagApB[[8]]
Out[*]=
10.98

In[*]:= ECMagApR[[8]]
Out[*]=
10.48

In[*]:= ECTeff[[8]]
Out[*]=
7244.9

In[*]:= Gcum[[256]]
Out[*]=
10.8008

In[*]:= Bcum[[256]]
Out[*]=
10.9863

In[*]:= Rcum[[256]]
Out[*]=
10.4774

In[*]:= Teffcum[[256]]
Out[*]=
7208.03

In[*]:= (*errores absolutos:*)

In[*]:= errAbsG = Norm[ECMagApG[[8]] - Gcum[[256]]]
          |norma
Out[*]=
0.00081363

In[*]:= errAbsB = Norm[ECMagApB[[8]] - Bcum[[256]]]
          |norma
Out[*]=
0.00627348

In[*]:= errAbsR = Norm[ECMagApR[[8]] - Rcum[[256]]]
          |norma
Out[*]=
0.00261766

In[*]:= errAbsTeff = Norm[ECTeff[[8]] - Teffcum[[256]]]
          |norma
Out[*]=
36.8698

```

```

In[*]:= (*errores relativos:*)

In[*]:= errRelG = errAbsG / EMagApG[[8]]
Out[*]:=
0.0000753361

In[*]:= errRelB = errAbsB / EMagApB[[8]]
Out[*]:=
0.000571355

In[*]:= errRelR = errAbsR / EMagApR[[8]]
Out[*]:=
0.000249777

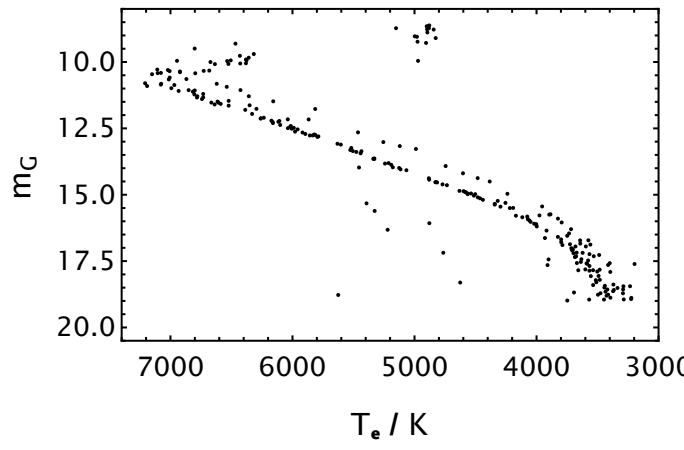
In[*]:= errRelTeff = errAbsTeff / ECTeff[[8]]
Out[*]:=
0.00508907

In[*]:= (*-----*)

In[*]:= (*APARTADO 8*)

In[*]:= ListPlot[Transpose[{Teffcum, Gcum}],
  represent... transposición
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  estilo base familia de tipo de letra tamaño de tipo de letra
  PlotRange → {{3000, 7400}, {8, 20.5}}, Frame → True,
  rango de representación marco verdadero
  FrameLabel → {Style["Te / K", 16, Bold], Style[Subscript[m, G], 16, Bold]},
  etiqueta de marco estilo negrita estilo subíndice negrita
  PlotStyle → {Black, Thickness[0.003]}, ScalingFunctions → {"Reverse", "Reverse"}]
  estilo de repre... negro grosor funciones de escalado invierte orden invierte orden
Out[*]:=

```



```

In[*]:= (*-----*)

In[*]:= (*APARTADO 9*)

In[*]:= HYDist = 1 / (HYParallax * 10^(-3)); (*el paralaje está en segundos de arco*)

```

```

In[*]:= HYDistMed = Mean[HYDist]
           [media]

Out[*]:=
46.7211

In[*]:= StandardDeviation[HYDist]
           [desviación estándar]

Out[*]:=
2.6245

In[*]:= HYMagAbsG = HYMagApG + 5 - 5 * Log10[HYDistMed];
           [logaritmo en base 10]

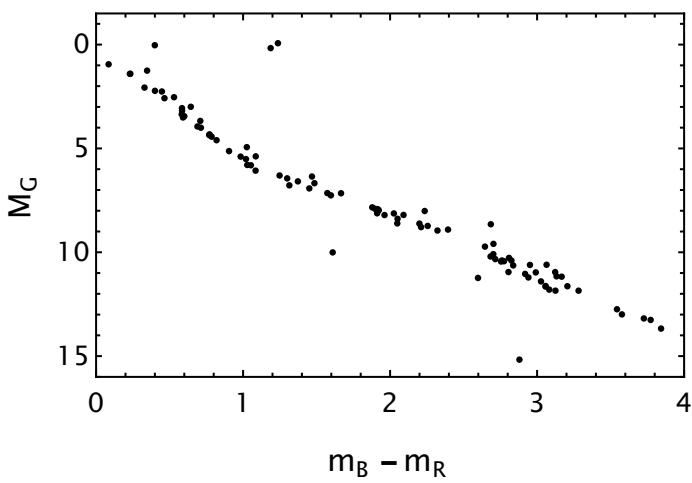
In[*]:= (*-----*)

In[*]:= (*APARTADO 10*)

In[*]:= BRhyad = HYMagApB - HYMagApR;

In[*]:= dataap10 = Transpose[{BRhyad, HYMagAbsG}];
           [transposición]

In[*]:= ListPlot[dataap10, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
           [representación de lista] [estilo base] [familia de tipo de letra] [tamaño de tipo de letra]
           PlotRange -> {{0, 4.0}, {-1.5, 16}}, Frame -> True, FrameLabel ->
           [rango de representación] [marco] [verd... [etiqueta de marco]
           {Style[Subscript[m, B] - Subscript[m, R], 16, Bold], Style[Subscript[M, G], 16, Bold]},
           [estilo] [subíndice] [subíndice] [negrita] [estilo] [subíndice] [negrita]
           PlotStyle -> {Black, Thickness[0.003]}, ScalingFunctions -> "Reverse"]
           [estilo de repre... [negro] [grosor] [funciones de escalado] [invierte orden]

Out[*]:=


```

In[*]:= (*-----*)

In[*]:= (*APARTADO 11*)

In[*]:= (*Hacemos coincidir las secuencias principales del
 cúmulo problema y del cúmulo de Hyades. Para ello utilizaremos
 G media del cúmulo problema y MG media del cúmulo de Hyades*)

```


```



```

In[*]:= HYMagAbsGMed = Mean[HYMagAbsG]
      media

Out[*]:=
7.59016

In[*]:= GcumMed = Mean[Gcum]
      media

Out[*]:=
14.1726

In[*]:= DistCum = 10^((GcumMed - HYMagAbsGMed) / 5 + 1) (*en pársecs*)

Out[*]:=
207.242

In[*]:= (*Lo de arriba no está del todo bien,
lo que hay que hacer es hacer coincidir las secuencias principales. Para ello hay que
hacer que MG sea igual en ambas. Así que busco un número que me las haga coincidir.*)

In[*]:= p1 = ListPlot[dataap10, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
      representación de lista  estilo base  familia de tipo de letra  tamaño de tipo de letra
      PlotRange -> {{0, 4.0}, {-1.5, 16}}, Frame -> True,
      rango de representación  marco  verdadero
      FrameLabel -> {Style[Subscript[m, B] - Subscript[m, R], 16, Bold],
      etiqueta de marco  estilo  subíndice  subíndice  negrita
      Style[Subscript[M, G], 16, Bold]},
      estilo  subíndice  negrita
      PlotStyle -> {Black, Thickness[0.003]}, ScalingFunctions -> "Reverse",
      estilo de repre...  negro  grosor  funciones de escalado  invierte orden
      PlotLegends -> Placed[PointLegend[{Black}, {"Cúmulo de Hyades"}], {Right, Top}]]];
      leyendas de rep...  colocado  punto leyenda  negro  derecha  arriba

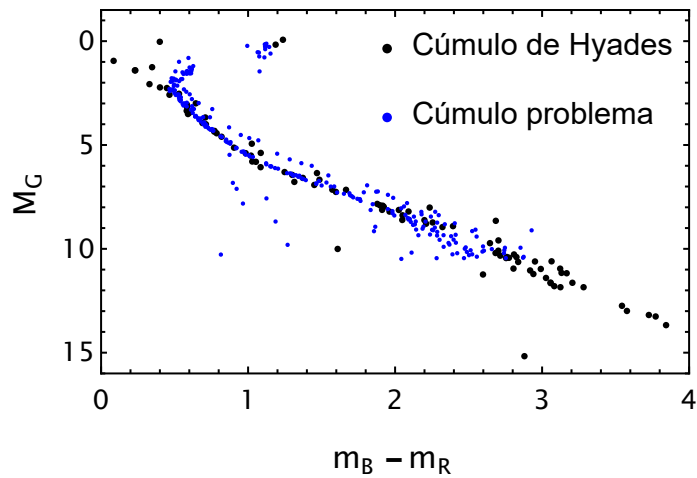
In[*]:= p2 = ListPlot[Transpose[{BRcumIntrins, Gcum - 8.5}],
      represent...  transposición
      BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
      estilo base  familia de tipo de letra  tamaño de tipo de letra
      PlotRange -> {{0, 4.0}, {-1.5, 16}}, Frame -> True,
      rango de representación  marco  verdadero
      FrameLabel -> {Style[Subscript[m, B] - Subscript[m, R], 16, Bold],
      etiqueta de marco  estilo  subíndice  subíndice  negrita
      Style[Subscript[M, G], 16, Bold]},
      estilo  subíndice  negrita
      PlotStyle -> {Blue, Thickness[0.003]}, ScalingFunctions -> "Reverse",
      estilo de repre...  azul  grosor  funciones de escalado  invierte orden
      PlotLegends -> Placed[PointLegend[{Blue}, {"Cúmulo problema"}], {Right, Top}]]];
      leyendas de rep...  colocado  punto leyenda  azul  derecha  arriba

```

```
In[*]:= Show[p1, p2]
```

[\[muestra\]](#)

```
Out[*]=
```



```
In[*]:= DistCumCorr = 10^(1 + (8.5 / 5)) (*en pc*)
```

```
Out[*]=
```

501.187

```
In[*]:= (*-----*)
          -----*)
```

```
In[*]:= (*APARTADO 12*)
```

```
In[*]:= MagAbsGcum = Gcum + 5 - 5 * Log10[DistCum];
```

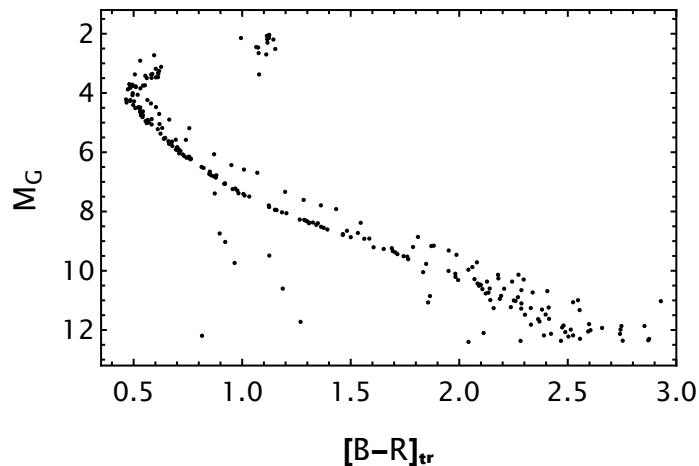
[\[logaritmo en base 10\]](#)

```
In[*]:= dataap12 = Transpose[{BRCumIntrins, MagAbsGcum}];
```

[\[transposición\]](#)

```
In[*]:= ListPlot[dataap12, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
  [representación de lista] [estilo base] [familia de tipo de letra] [tamaño de tipo de letra]
  PlotRange -> {{0.35, 3.0}, {1.2, 13.2}}, Frame -> True,
  [rango de representación] [marco] [verdadero]
  FrameLabel -> {Style["[B-R]tr", 16, Bold], Style[Subscript[M, G], 16, Bold]},
  [etiqueta de marco] [estilo] [negrita] [estilo] [subíndice] [negrita]
  PlotStyle -> {Black, Thickness[0.003]}, ScalingFunctions -> "Reverse"]
  [estilo de repre...] [negro] [grosor] [funciones de escalado] [invierte orden]
```

```
Out[*]=
```



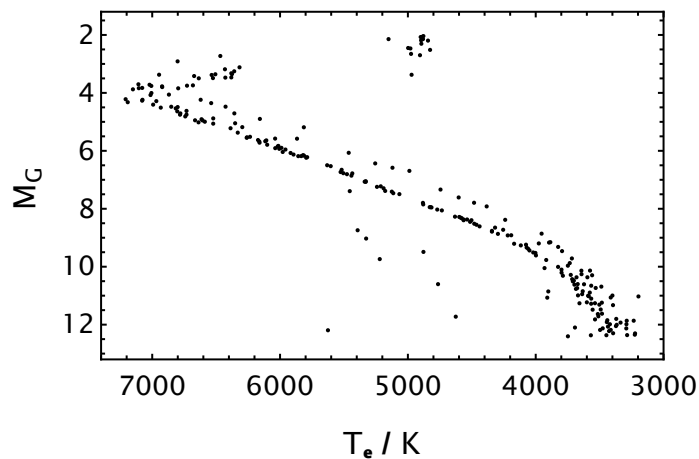
```
In[*]:= (*-----*)
                      -----*)
```

```
In[*]:= (*APARTADO 8*) (*diagrama HR del cúmulo problema*)
```

```
In[*]:= datap8 = Transpose[{Teffcum, MagAbsGcum}];
           |transposición
```

```
In[*]:= ListPlot[datap8, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
           |representación de lista |estilo base |familia de tipo de letra |tamaño de tipo de letra
           PlotRange → {{3000, 7400}, {1.2, 13.2}}, Frame → True,
           |rango de representación |marco |verdadero
           FrameLabel → {Style["Te / K", 16, Bold], Style[Subscript[M, G], 16, Bold]},
           |etiqueta de marco |estilo |negrita |estilo |subíndice |negrita
           PlotStyle → {Black, Thickness[0.003]}, ScalingFunctions → {"Reverse", "Reverse"}]
           |estilo de repre... |negro |grosor |funciones de escalado |invierte orden |invierte orden
```

```
Out[*]:=
```



```
In[*]:= (*-----*)
                      -----*)
```

```
In[*]:= (*APARTADO 13*)
```

```
In[*]:= (*-----*)
                      -----*)
```

```
In[*]:= (*APARTADO 14*)
```

```
In[*]:= BVJohn = BRcumIntrins + 0.12;
```

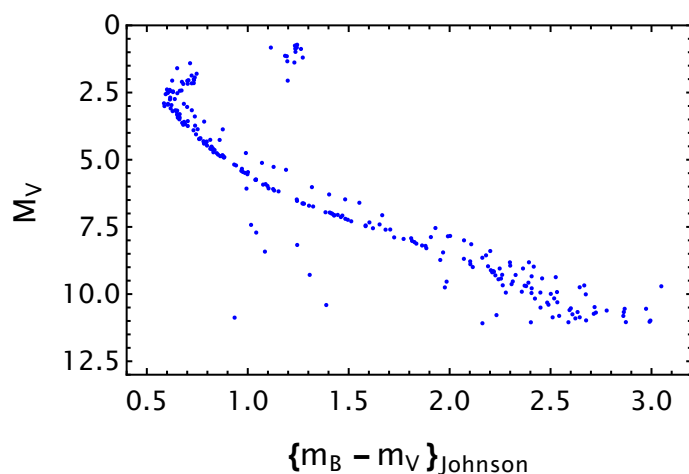
```
In[*]:= MVCum = (Gcum - 8.5) + 0.6; (*Magnitud Absoluta G del cúmulo = Gcum - 8.5*)
```

```

In[*]:= ListPlot[Transpose[{BVJohn, MVCum}],
  represent... [transposición]
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  [estilo base] [familia de tipo de letra] [tamaño de tipo de letra]
  PlotRange → {{0.4, 3.2}, {0, 13}}, Frame → True,
  [rango de representación] [marco] [verdadero]
  FrameLabel → {Style[Subscript[{Subscript[m, B] - Subscript[m, V]}, Johnson], 16, Bold],
  [etiqueta de marco] [estilo] [subíndice] [subíndice] [subíndice] [negrita]
  Style[Subscript[M, V], 16, Bold]},
  [estilo] [subíndice] [negrita]
  PlotStyle → {Blue, Thickness[0.003]}, ScalingFunctions → "Reverse" (*escalado*)
  [estilo de repre...] [azul] [grosor] [funciones de escalado] [invierte orden]

```

Out[*]=



```

In[*]:= ListPlot[Transpose[{BVJohn, MVCum}],
  represent... [transposición]
  BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  [estilo base] [familia de tipo de letra] [tamaño de tipo de letra]
  PlotRange → {{-0.3, 2.2}, {-8, 8}}, Frame → True,
  [rango de representación] [marco] [verdadero]
  FrameLabel → {Style[Subscript[{Subscript[m, B] - Subscript[m, V]}, Johnson], 16, Bold],
  [etiqueta de marco] [estilo] [subíndice] [subíndice] [subíndice] [negrita]
  Style[Subscript[M, V], 16, Bold]}, PlotStyle → {Blue, Thickness[0.003]},
  [estilo] [subíndice] [negrita] [estilo de repre...] [azul] [grosor]
  ScalingFunctions → "Reverse", PlotMarkers → {Automatic, Medium}]
  [funciones de escalado] [invierte or...] [marcadores de re...] [automático] [tamaño medio]
  (*a escala de la fig 7.10 de la web*)

```

Out[*]=

